

CV2090

Probability & Statistics for Civil Engineers

(Jul – Nov 2026)

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L14
Aug 24, 2026 (8 am)

Announcement

- 24th Aug (today):
 - Lecture hour 3 – 4 pm
 - Tutorial-4: 4 – 5 pm
- 25th Aug: No class on Tuesday
- 31st Aug: Quiz-1 (8 AM)
 - Quiz-1 syllabus: up to random variables (including Normal and χ^2 , but excluding functions of rv)
- Check course page on GitHub regularly
<https://github.com/prakashvs613/2026-CV2090-IITM/>

Normal/Gaussian Distribution

Ang and Tang (Section 3.2.1)

The Normal Distribution

Motivation → Equation → Standardization/Z-table →
reading table/reverse look-up

Normal distribution: motivation

→ human height

- genetics → nutrition during first 1000's of days
- skeleton response to mechanical stress

- all of these RVs follow Normal Distribution

- Most ubiquitous distribution

- Several Additive effects

- Mathematically Tractable.

NORMAL / GAUSSIAN /
BELL CURVE

→ Concrete strength

- Aggregate sizes → exact water-to-cement ratio
- chemical impurities → microscopic air traps

→ Standardized Test Scores

- Preparation → right before
- 50/50 → mnemonic devices

Normal (Gaussian) PDF

PDF

$$f_X(x) = \frac{1}{\sqrt{2\pi} \sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

$$f_X(x; \mu, \sigma)$$

↑
parameters

$$-\infty < x < \infty$$

CDF does NOT have a closed-form solution.

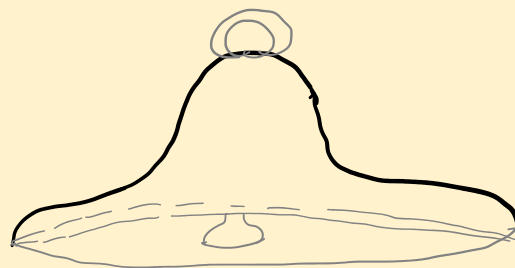
$$E[X] = \mu$$

$$\text{Var}[X] = \sigma^2$$

$$\theta[X] = 0$$

skewness

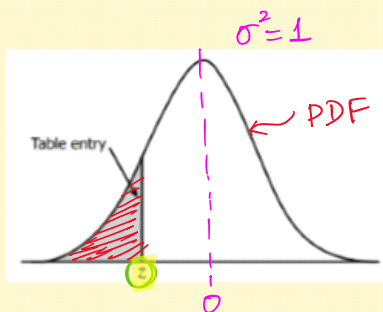
$$\kappa[X] = 3$$



7

Standard Normal Probabilities

Z-table

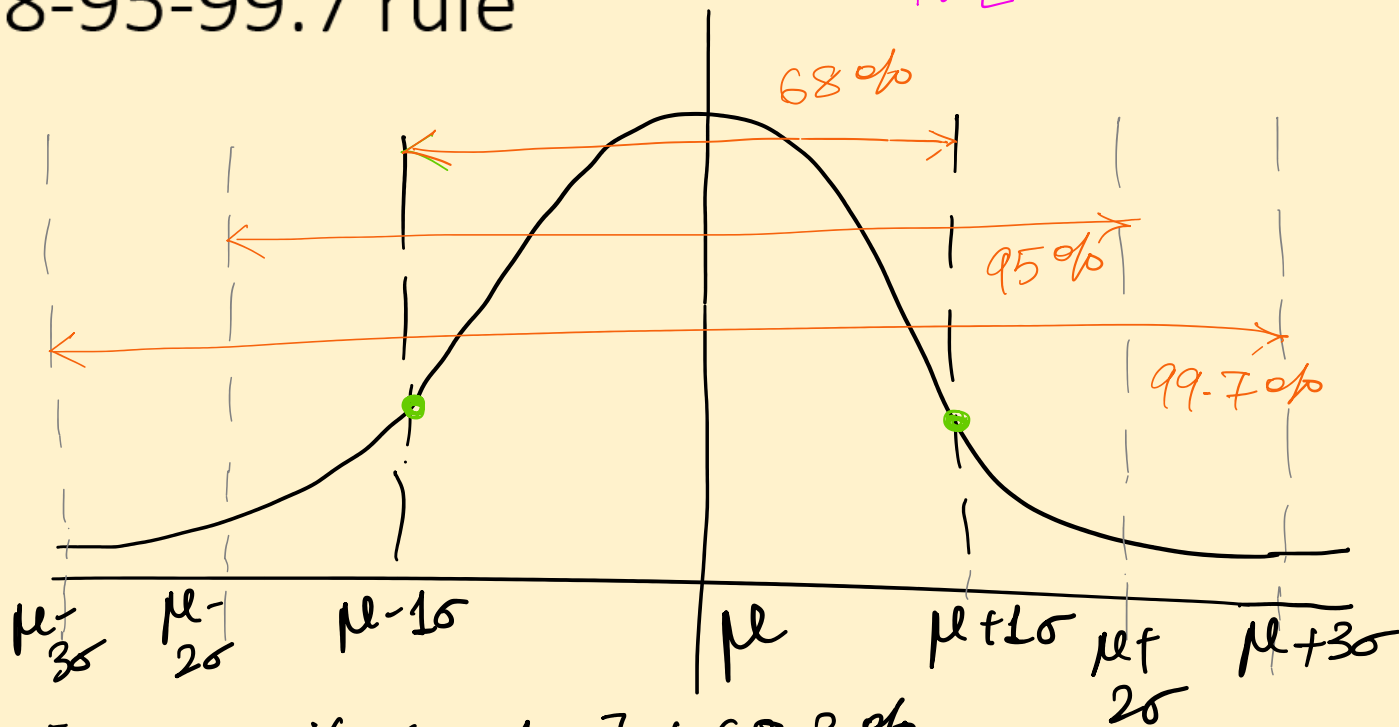


z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
-3.4	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0002
-3.3	.0005	.0005	.0005	.0004	.0004	.0004	.0004	.0004	.0004	.0003
-3.2	.0007	.0007	.0006	.0006	.0006	.0006	.0006	.0005	.0005	.0005
-3.1	.0010	.0009	.0009	.0009	.0008	.0008	.0008	.0008	.0007	.0007
-3.0	.0013	.0013	.0013	.0012	.0012	.0011	.0011	.0011	.0010	.0010
-2.9	.0019	.0018	.0018	.0017	.0016	.0016	.0015	.0015	.0014	.0014
-2.8	.0026	.0025	.0024	.0023	.0023	.0022	.0021	.0021	.0020	.0019
-2.7	.0035	.0034	.0033	.0032	.0031	.0030	.0029	.0028	.0027	.0026
-2.6	.0047	.0045	.0044	.0043	.0041	.0040	.0039	.0038	.0037	.0036
-2.5	.0062	.0060	.0059	.0057	.0055	.0054	.0052	.0051	.0049	.0048
-2.4	.0082	.0080	.0078	.0075	.0073	.0071	.0069	.0068	.0066	.0064
-2.3	.0107	.0104	.0102	.0099	.0096	.0094	.0091	.0089	.0087	.0084
-2.2	.0139	.0136	.0132	.0129	.0125	.0122	.0119	.0116	.0113	.0110
-2.1	.0179	.0174	.0170	.0166	.0162	.0158	.0154	.0150	.0146	.0143
-2.0	.0228	.0222	.0217	.0212	.0207	.0202	.0197	.0192	.0188	.0183
-1.9	.0287	.0281	.0274	.0268	.0262	.0256	.0250	.0244	.0239	.0233
-1.8	.0359	.0351	.0344	.0336	.0329	.0322	.0314	.0307	.0301	.0294
-1.7	.0446	.0436	.0427	.0418	.0409	.0401	.0392	.0384	.0375	.0367
-1.6	.0548	.0537	.0526	.0516	.0505	.0495	.0485	.0475	.0465	.0455
-1.5	.0668	.0655	.0643	.0630	.0618	.0606	.0594	.0582	.0571	.0559
-1.4	.0808	.0793	.0778	.0764	.0749	.0735	.0721	.0708	.0694	.0681
-1.3	.0968	.0951	.0934	.0918	.0901	.0885	.0869	.0853	.0838	.0823
-1.2	.1151	.1131	.1112	.1093	.1075	.1056	.1038	.1020	.1003	.0985
-1.1	.1357	.1335	.1314	.1292	.1271	.1251	.1230	.1210	.1190	.1170
-1.0	.1587	.1562	.1539	.1515	.1493	.1472	.1450	.1429	.1408	.1388

Table entry for z is the area under the standard normal curve to the left of z .

-1.3	.0968	.0951	.0934	.0918	.0901	.0885	.0869	.0853	.0838	.0823
-1.2	.1151	.1131	.1112	.1093	.1075	.1056	.1038	.1020	.1003	.0985
-1.1	.1357	.1335	.1314	.1292	.1271	.1251	.1230	.1210	.1190	.1170
-1.0	.1587	.1562	.1539	.1515	.1492	.1469	.1446	.1423	.1401	.1379
-0.9	.1841	.1814	.1788	.1762	.1736	.1711	.1685	.1660	.1635	.1611
-0.8	.2119	.2090	.2061	.2033	.2005	.1977	.1949	.1922	.1894	.1867
-0.7	.2420	.2389	.2358	.2327	.2296	.2266	.2236	.2206	.2177	.2148
-0.6	.2743	.2709	.2676	.2643	.2611	.2578	.2546	.2514	.2483	.2451
-0.5	.3085	.3050	.3015	.2981	.2946	.2912	.2877	.2843	.2810	.2776
-0.4	.3446	.3409	.3372	.3336	.3300	.3264	.3228	.3192	.3156	.3121
-0.3	.3821	.3783	.3745	.3707	.3669	.3632	.3594	.3557	.3520	.3483
-0.2	.4207	.4168	.4129	.4090	.4052	.4013	.3974	.3936	.3897	.3859
-0.1	.4602	.4562	.4522	.4483	.4443	.4404	.4364	.4325	.4286	.4247
-0.0	.5000	.4960	.4920	.4880	.4840	.4801	.4761	.4721	.4681	.4641

68-95-99.7 rule



$$Pr[\mu - 1\sigma \leq X \leq \mu + 1\sigma] \approx 68.3\%$$

$$Pr[\mu - 2\sigma \leq X \leq \mu + 2\sigma] \approx 95.4\%$$

$$Pr[\mu - 3\sigma \leq X \leq \mu + 3\sigma] \approx 99.7\%$$

Features of Normal Distributions

- Symmetric $\Rightarrow$ mean = median = mode
- Smooth, with 2 points of inflection (@ $\mu \pm 1\sigma$)
- Completely defined by

→ mean, μ

→ variance, σ^2

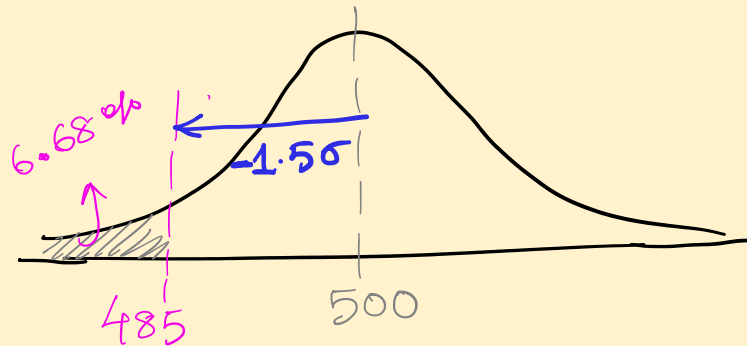
$$\rightarrow \Pr(X \in [\mu - 1\sigma, \mu + 1\sigma]) \approx 68\%$$

$$-2\sigma, \quad +2\sigma \quad \approx 95\%$$

$$-3\sigma, \quad +3\sigma \quad \approx 99.7\%$$

Class problems

- **Example 1:** Given a factory machine filling cereal boxes with a mean weight $\mu = 500$ g and $\sigma = 10$ g, find the probability that a randomly chosen box weighs less than 485 g



- **Example 2:** Find the threshold weight that marks the top 10% of heaviest boxes.

